



Tennessee Valley Authority, Post Office Box 2000, Spring City, Tennessee 37381

August 8, 2016

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Watts Bar Nuclear Plant, Units 1 and 2
Facility Operating License Nos. NPF-90 and 96
NRC Docket Nos. 50-390 and 50-391

Subject: **Licensee Event Report 50-390 and 50-391/2016-010-00, Emergency Diesel Generator Crankcase Pressure Switches Not Analyzed to Withstand the Effects of a Tornado**

This submittal provides Licensee Event Report (LER) 390 and 391/2016-010-00. This LER provides details concerning a vulnerability in the design of the Emergency Diesel Generator (EDG) that may result in an EDG lockout prior to emergency start during a tornado. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(ii)(B).

Please direct any questions concerning this matter to Gordon Arent, WBN Licensing Director, at (423) 365-2004.

Respectfully,

A handwritten signature in black ink, appearing to read "Paul Simmons", with a long horizontal flourish extending to the right.

Paul Simmons
Site Vice President
Watts Bar Nuclear Plant

Enclosure
cc: See Page 2

U.S. Nuclear Regulatory Commission
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cc (Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Watts Bar Nuclear Plant



LICENSEE EVENT REPORT (LER)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by internet e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME

Watts Bar Nuclear Plant, Unit 1

2. DOCKET NUMBER

05000390

3. PAGE

1 OF 5

4. TITLE

Emergency Diesel Generator Crankcase Pressure Switches Not Analyzed to Withstand the Effects of a Tornado

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
06	08	2016	2016	- 010	- 00	08	08	2016	Watts Bar Nuclear Plant Unit 2	05000391
									FACILITY NAME	DOCKET NUMBER
									N/A	N/A

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☒ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

Dean Baker, Licensing Engineer

TELEPHONE NUMBER (Include Area Code)

423-452-4589

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX

14. SUPPLEMENTAL REPORT EXPECTED

☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO

15. EXPECTED SUBMISSION DATE

MONTH	DAY	YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

At 1526 Eastern Daylight Time on 6/8/2016, a determination was made involving the potential impact of a tornado on the Emergency Diesel Generators (EDGs). The EDGs are required to be operable to provide power to ensure that acceptable fuel design limits, reactor coolant system pressure boundary limits, and containment integrity are not exceeded during abnormal transients. Further, the EDGs are designed with a crankcase pressure trip (setpoint of 1 inch water), which is bypassed following an emergency start. Engineering has determined that a tornado could potentially cause actuation of the crankcase pressure trip due to a low barometric condition. If an emergency start signal has NOT previously occurred, then during a tornado, actuation of the crankcase pressure trip would energize the shutdown relay causing an EDG lockout condition. The EDG lockout condition prevents subsequent EDG starts (normal or emergency) until operators manually reset the lockout condition locally at the EDG. This condition could potentially affect all four EDGs simultaneously.

A compensatory measure has been established, that upon notification of a Tornado Warning, the EDGs would be 'emergency started' and run during the time the Tornado Warning was in effect. This action bypasses the crankcase pressure trip function and allows the EDGs to perform their required safety function.

NRC FORM 366A
(11-2015)

U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104

EXPIRES: 10/31/2018



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Watts Bar Nuclear Plant, Unit 1	05000390	YEAR	SEQUENTIAL NUMBER	REV NO.
		2016	- 010	- 00

NARRATIVE

I. PLANT OPERATING CONDITIONS BEFORE THE EVENT

Watts Bar Nuclear Plant (WBN) Unit 1 was in Mode 1 at 100 percent rated thermal power (RTP) and WBN Unit 2 was in Mode 1 at 8 percent RTP.

II. DESCRIPTION OF EVENT

A. Event

At 1526 Eastern Daylight Time (EDT) on June 8, 2016, a determination was made involving the potential impact of a tornado on the Emergency Diesel Generators (EDGs) {EIS:DG}. The EDGs are required to be operable to provide power to ensure that acceptable fuel design limits, reactor coolant system pressure boundary limits, and containment integrity are not exceeded during abnormal transients. Further, the EDGs are designed with a crankcase pressure trip (setpoint of 1 inch water), which is bypassed following an emergency start. Engineering has determined that a tornado could potentially cause actuation of the crankcase pressure trip due to a low barometric condition. If an emergency start signal has NOT previously occurred, then during a tornado, actuation of the crankcase pressure trip would energize the shutdown relay causing an EDG lockout condition. The EDG lockout condition prevents subsequent EDG starts (normal or emergency) until operators manually reset the lockout condition locally at the EDG. This condition could potentially affect all four EDGs simultaneously.

The EDGs are operable but degraded. All EDGs have successfully passed their required surveillances within the appropriate frequency. No severe weather warnings or watches are forecast in the local areas, which could challenge the crankcase pressure trip.

A compensatory measure has been established, that upon notification of a Tornado Warning, the EDGs would be 'emergency started' and run during the time the Tornado Warning was in effect. This action bypasses the crankcase pressure trip function and allows the EDGs to perform their required safety function.

The identified condition placed both units in an unanalyzed condition that potentially significantly degrades plant safety. This event was reported to the Nuclear Regulatory Commission (NRC) on June 8, 2016 via Event Notification 51995 pursuant to 10 CFR 50.72(b)(2)(ii)(B). It is being reported in this LER as an unanalyzed condition pursuant to 10 CFR 50.73(a)(2)(ii)(B).

B. Inoperable Structures, Components, or Systems that Contributed to the Event

No inoperable structures, systems, or components contributed to this event.

C. Dates and Approximate Times of Occurrences

Date	Time (EDT)	Event
6/7/16		Condition Report (CR) 1179264 initiated for WBN based on an issue identified at the Sequoyah Nuclear Plant related to a design vulnerability of the EDGs during a tornado.

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(11-2015)

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Date	Time (EDT)	Event
6/8/16	1526	WBN reports the above condition for the crankcase over-pressure switches to the NRC.

D. Manufacturer and Model Number of Components that Failed

Not applicable.

E. Other Systems or Secondary Functions Affected

No other systems or secondary functions were affected.

F. Method of discovery of each Component or System Failure or Procedural Error

This unanalyzed condition was discovered at the Sequoyah Nuclear Plant (SQN) during an NRC Component Design Basis Inspection. Because WBN and SQN are essentially identical designs, this issue was promptly reviewed for applicability at Watts Bar.

G. Failure Mode and Effect of Each Failed Component

Not applicable.

H. Operator Actions

Upon determination of the unanalyzed condition, the NRC was promptly notified.

I. Automatically and Manually Initiated Safety System Responses

There were no automatic or manual system responses associated with this event.

III. CAUSE OF THE EVENT

A. The cause of each component or system failure or personnel error, if known.

The design of the interlock circuit utilizes the pressure between the crankcase and the room was created more than 20 years ago. It is not possible to identify the specific individuals involved in the design to determine their mindset at the time this design was developed. Therefore, the cause of the event is considered a latent or legacy issue, and not a weakness in the current design process.

B. The cause(s) and circumstances for each human performance related root cause.

The issue of the interlock design is a latent condition that was created more than 20 years ago.

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IV. ANALYSIS OF THE EVENT

All four EDGs are required to be operable by both units' Technical Specifications to provide electrical power to safe shutdown/safety related equipment following accident conditions coincident with a loss of offsite power. The Current Licensing Basis (CLB) requires that tornado effects be considered in the design of safety related structures, systems, and components. It could not be demonstrated at the time of the event that the crankcase pressure switches would withstand the design basis tornado.

The EDGs are operable, but degraded and have successfully passed their required surveillances within the appropriate frequency. The occurrence of such an event involving failure of the crankcase pressure switches is highly unlikely and there was no imminent concern regarding severe weather involving tornadoes.

In order to prevent any operability issues with the EDG due to tornado damage to the EDG building or components, compensatory measures have been developed to address the associated nonconformance. Upon notification of a Tornado Warning, the EDGs will be emergency started and operated during the time the Tornado Warning is in effect to bypass the crankcase pressure trip function and allow the EDGs to be able to perform their required safety function.

V. ASSESSMENT OF SAFETY CONSEQUENCES

The EDGs are located inside the power plant structure and are capable of performing their safety function. The occurrence of such an event involving failure of the crankcase pressure switches is highly unlikely and there was no imminent concern regarding severe weather involving tornadoes at the time of this event.

A risk assessment has been completed and has determined that the change in core damage frequency is less than 1E-6 for failure of the EDG crankcase pressure switches, which is of very low safety significance.

- A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

There were no component failures .

- B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

These events did not occur when the reactor was shutdown.

- C. For failure that rendered a train of a safety system inoperable, an estimate of the elapsed time from the discovery of the failure until the train was returned to service

With the compensatory measures in place, the EDGs will remain capable of fulfilling their design basis function following a tornado.

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NARRATIVE

VI. CORRECTIVE ACTIONS

This event was entered into the Tennessee Valley Authority Corrective Action Program and is being tracked under CR 1179264.

A. Immediate Corrective Actions

An 8-hour non-emergency notification was made to the NRC to report the event as an unanalyzed condition that significantly degrades plant safety per 10 CFR 50.72(b)(3)(ii)(B).

B. Corrective Actions to Prevent Recurrence

Corrective actions include establishing a compensatory measure that upon notification of a Tornado Warning, the EDGs will be emergency started and operated during the time the Tornado Warning is in effect to bypass the crankcase pressure trip function and allow the EDGs to perform their required safety function.

A design change to remove the crankcase pressure trip is being developed and is expected to be completed in December 2016. Implementation of the modification will then be scheduled during an appropriate work period for each EDG.

VII. ADDITIONAL INFORMATION

A. Previous similar events at the same plant

No similar events have been identified at the Watts Bar plant.

B. Additional Information

None.

C. Safety System Functional Failure Consideration

This condition did not result in a safety system functional failure.

D. Scrams with Complications Consideration

There were no scrams associated with this event.

VIII. COMMITMENTS

None.